VISTULA UNIVERSITY

The Faculty of Computer Science, Graphics and Architecture

Program of study Computer Science

Inji Ahmadzada

Student's number 51516

IMPLEMENTATION OF MACHINE LEARNING ALGORITHMS IN CRYPTOANALYSIS

The engineering thesis

written under the supervision of

Dr. Hab. Maciej Janowicz

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Abstract

The use of machine learning algorithms in cryptanalysis has emerged as a promising area of research in recent years. My thesis aims to explore the implementation of machine learning algorithms for cryptanalysis, specifically in the context of the Vigenère cipher. The proposed method involves generating a random plaintext and key, encrypting the plaintext using the Vigenère cipher, and then using a neural network to perform a frequency analysis attack on the ciphertext and recover the original plaintext. My thesis also discusses the limitations of existing methods for human action recognition, which typically use hand-crafted features and suffer from limited discriminative power and sensitivity to variations in the input data. The proposed method demonstrates the effectiveness of deep learning algorithms for action recognition and provides valuable insights into the use of these algorithms for complex and challenging tasks in computer vision. Overall, the results show that machine learning algorithms can be successfully applied to cryptanalysis and have the potential to significantly improve the security of cryptographic systems.

Key words: machine learning, algorithms, cryptography, cryptanalysis, ciphers, decryption, cryptography, neural networks.

Chapter 1 – Introduction

1.1. Why this topic?

As we know technology is constantly developing and growing to make our lives convenient and practical. Machine Learning, one of the remarkable areas of IT, has gained lots of attention and popularity in the last few years and keeps increasing its use in the tech world. Due to this, I wanted to dedicate my thesis to Machine Learning implementation. However, I had an abundant interest in another topic, which is called Cryptanalysis, in particular the decryption of ciphers. After some research and consideration, I decided to somehow combine these topics and illustrate my approach to the integration of Machine Learning and Cryptanalysis.

In my research, I discovered there were very few efficient ways to decipher the various ciphers. This is because it is an old, simple system. Unfortunately, it takes a lot of time to achieve any result. During my research I bumped into different use cases and research from previous years and got inspiration for my paper. I will explain this in more detail in section 1.2.

1.2. Aims and Objectives

My objective is to demonstrate the use of machine learning techniques, specifically neural networks, for cryptanalysis by attacking the Vigenère cipher. The aim of the project that will support my thesis is to generate a random plaintext and key, encrypt the plaintext using the Vigenère cipher, and then use a neural network to perform a frequency analysis attack on the ciphertext and recover the original plaintext. This will reduce time consumption, errors, confusion, and machine breakdowns. The project also demonstrates the principles of feature engineering, training, and testing a machine learning model, and using the model to make predictions on new data. For my project, I am using python programming language, learning models, different libraries, and data sets to achieve my end goal. Mainly, my program and thesis will demonstrate the

integration of Machine Learning algorithms in Cryptography, particularly in Cryptanalysis.

1.3. Structure Outline

The following chapters describe the general idea of Machine Learning and Machine Learning algorithms and their usage in various circumstances. Additionally, information relating to Cryptology, Cryptanalysis, and Ciphers and their decryption techniques are defined accordingly. Next, I describe the implementation of Machine Learning algorithms in Cryptanalysis with detailed explanations and supporting use cases. Later, I talk about why I chose to create “Decrypted” and the tools I used for it with brief explanations and illustrations also the documentation of how the code works. Finally, the last chapter summarizes the whole project and gives some information about future implementation ideas.

Chapter 2 - Machine Learning

2.1. What is Machine Learning?

Developing techniques and models that allow computers to learn from data without being explicitly programmed is the focus of the rapidly expanding discipline of machine learning (ML). The fundamental concept underlying machine learning (ML) is to give computers the capacity to learn from data and enhance their performance over time without requiring human involvement. In order to do this, statistical and mathematical models are used. These models may find patterns in the data, forecast the future based on those patterns, and constantly improve as more data becomes available.

One of machine learning's key characteristics is its capacity to provide customers answers that are difficult to get using conventional techniques. For instance, ML may be employed to examine big datasets and spot patterns that are difficult for people to notice. This may result in fresh perceptions and learnings that may be applied to better client experiences, create new products, and boost corporate operations.[1]

Machine learning's capacity for handling vast volumes of data is another crucial aspect. ML approaches are becoming more and more crucial for processing and interpreting this data as the volume of data produced by people and companies keeps rising. Large datasets may be analyzed using ML algorithms to find patterns, trends, and links that help firms make better decisions and plan their strategy.

Additionally, ML is employed in a variety of fields, including business process automation (BPA), predictive maintenance, email filtering, recommendation systems, fraud detection, spam filtering, and malware threat identification. Data analysis, forecasting, and performance enhancement are all achieved in each of these fields using machine learning (ML).[2]

[1] Alpaydin, E. (2010). Introduction to machine learning (2nd ed.). Cambridge, MA: MIT Press.https://kkpatel7.files.wordpress.com/2015/04/alppaydin_machinelearning_2010.pdf (accessed on January 9, 2023)

[2] Jordan, M. I., & Mitchell, T. M. (2015). Machine learning: Trends, perspectives, and prospects. Science, 349(6245), 255-260.https://www.cs.cmu.edu/~tom/pubs/Science-ML-2015.pdf (Accessed on January 9, 2023)

The capacity of machine learning (ML) to uncover hidden patterns in data is one of its main advantages. This is achieved through the application of sophisticated mathematical models, which are able to spot connections and correlations that could escape the notice of human analysts. New understandings and discoveries may result from this, which may then be used to the creation of fresh goods, services, and business plans.

Overall, machine learning is an effective technology that is quickly changing how companies and organizations conduct their operations. ML is able to offer insights and solutions that were previously unachievable by utilizing the power of data and algorithms. In the upcoming years, it is possible that we will witness even more major advancements in the sector as more data becomes accessible and ML approaches continue to grow. [3]

I would like to highlight the main features of machine learning:

1. Produce solutions for consumers that are challenging to obtain.
2. Quick increment in data production
3. Making decisions in various domains.
4. Extraction of hidden patterns from the provided data by users.
5. It is a data-driven technology.
6. It deals with a huge amount of data like data mining.

2.2. Machine Learning Algorithms

Machine learning algorithms are a group of mathematical models that are used to data analysis and learning in order to produce predictions or judgments. These algorithms may be divided into a number of categories, such as **reinforcement learning**, **unsupervised learning**, **semi-supervised learning**, and **supervised learning**.[4]

[3] Bishop, C. M. (2006). Pattern recognition and machine learning (1st ed.). New York: Springer.http://users.isr.ist.utl.pt/~wurmd/Livros/school/Bishop%20-%20Pattern%20Recognition%20And%20Machine%20Learning%20-%20Springer%20%202006.pdf (Accessed on January 9, 2023)

[4] Mitchell, T. (1997). Machine learning. Boston, MA: McGraw-Hill. https://www.cin.ufpe.br/~cavmj/Machine%20-%20Learning%20-%20Tom%20Mitchell.pdf (Accessed on January 23, 2023)

2.2.1. Supervised Learning

Labeled datasets are used in *supervised machine learning* to develop algorithms for accurately classifying data or predicting outcomes. It assists enterprises in finding scalable solutions to a range of real-world issues, such as categorizing spam in a different folder from your inbox. When training and processing are complete, the system builds a model using labeled data to interpret the datasets and learn about each one. After the model has finished, we test it using sample data to see if it accurately predicts the intended result. The integration of input and output data is the aim of supervised learning. Techniques used in supervised learning include neural networks, linear regression, logistic regression, random forest, and support vector machines (SVM). Example algorithms include *Logistic Regression* and the *Back Propagation Neural Network*.

- **Linear regression**

  Linear regression is a statistical regression method used for predictive analysis and shows the linear relationship between the independent variable (X-axis) and the dependent variable (Y-axis).

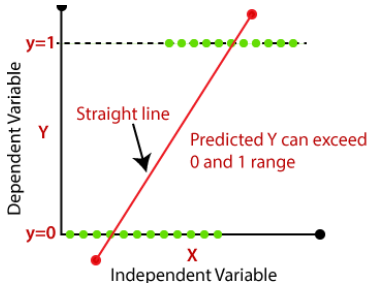


Figure 1. Linear regression visual description. https://www.javatpoint.com/linear-regression-vs-logistic-regression-in-machine-learning

- **Logistic regression**

  Logistic regression is a statistical data analysis method that employs mathematics to find the interactions between two data factors. Data and the relationship between one dependent variable and one or more independent variables are described via logistic regression. Nominal, ordinal, or interval types can all be used for the independent variables. The value of one of those elements is then predicted depending on the other

using this connection. The outcome of the forecast often has a limited range, such as yes or no. [5]

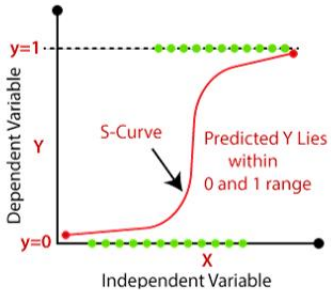


Figure 2. Logistic regression visual description. *https://www.javatpoint.com/linear-regression-vs-logistic-regression-in-machine-learning*

- **Decision tree**

A decision tree is a non-parametric supervised learning algorithm used for classification and regression tasks. It has a hierarchical, tree structure, which contains a root node, branches, internal nodes, and leaf nodes.[6]

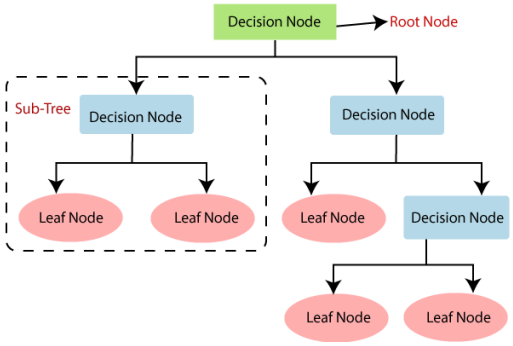


Figure 3. Decision tree visual description. *https://www.javatpoint.com/machine-learning-decision-tree-classification-algorithm*

[5] What is Logistic Regression? - Logistic Regression Model Explained - AWS. (n.d.). Amazon Web Services, Inc. https://aws.amazon.com/what-is/logistic-regression/ (Accessed on January 29, 2023)
[6] What is a Decision Tree | IBM. (n.d.). What Is a Decision Tree | IBM. https://www.ibm.com/topics/decision-trees (Accessed on January 29, 2023)

- **SVM algorithm (Support Vector Machine Algorithm)**

  This algorithm is used for classification and regression. Finding a hyperplane in an N-dimensional space that classifies the data points is the goal of the SVM method. The number of features determines the hyperplane's size. If there are just two input features, the hyperplane is merely a line; however, if there are three input characteristics, the hyperplane transforms into a two-dimensional plane. It is exceedingly difficult to move forward when there are more than three, as you may assume.[7]

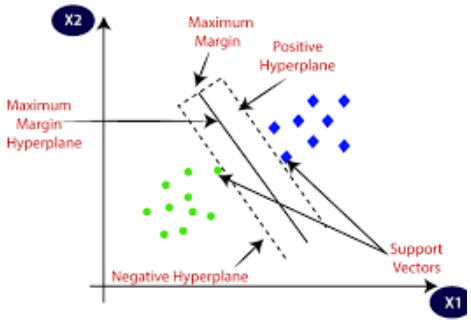


Figure 4. SVM algorithm visual representation. *https://www.javatpoint.com/machine-learning-support-vector-machine-algorithm*

- **Naïve Bayes algorithm**

  Naïve Bayes assumes conditional independence over the training dataset. Data are divided into distinct classes by the classifier in accordance with Bayes' Theorem. Based on past knowledge of the circumstances that could be relevant to the event, Bayes' Theorem estimates the likelihood of an event.

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}$$

Figure 5. Naïve Bayes theorem. *https://www.naukri.com/learning/articles/naive-bayes-algorithm-in-machine-learning/#nba*

[7] *Support Vector Machine Algorithm - GeeksforGeeks*. (2021, January 20). GeeksforGeeks. https://www.geeksforgeeks.org/support-vector-machine-algorithm/ (Accessed on January 31, 2023)

However, assumes that the interaction between each input feature in a class is unrelated. Thus, the model is said to as naïve.[8]

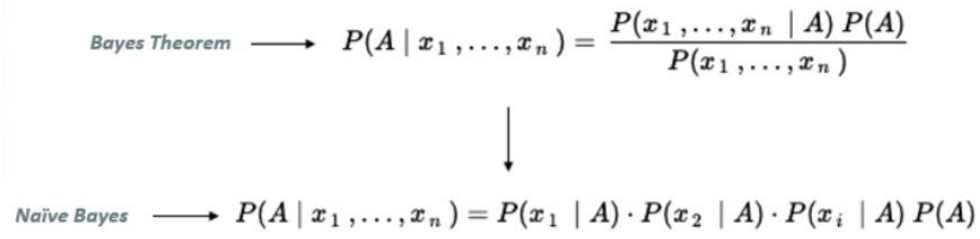


Figure 6. Naïve Bayes theorem and algorithm visual representation.
https://www.naukri.com/learning/articles/naive-bayes-algorithm-in-machine-learning/#nba

- **Random forest algorithm**

On various samples, it constructs decision trees and uses their average for classification and majority vote for regression.

The Random Forest Algorithm's ability to handle data sets with both continuous variables, as in regression, and categorical variables, as in classification, is one of its most crucial qualities. In terms of categorization issues, it delivers superior outcomes.[9]

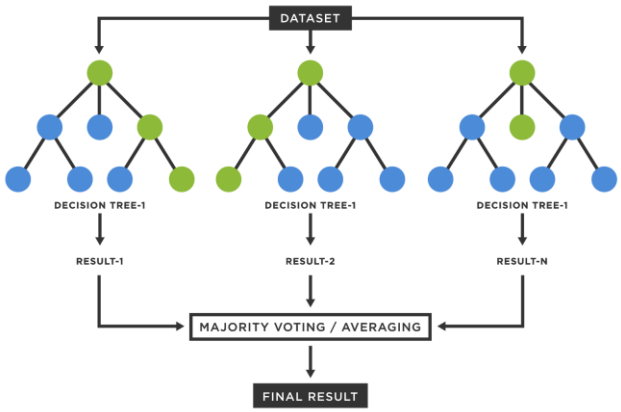


Figure 7. Random forest algorithm visual representation.
https://www.tibco.com/reference-center/what-is-a-random-forest

[8] https://www.naukri.com/learning/articles/naive-bayes-algorithm-in-machine-learning/#nba. (n.d.). https://www.naukri.com/learning/articles/naive-bayes-algorithm-in-machine-learning/#nba (Accessed on January 31, 2023)

[9] R, S. E. (2021, June 17). Random Forest Algorithms - Comprehensive Guide With Examples. Analytics Vidhya. https://www.analyticsvidhya.com/blog/2021/06/understanding-random-forest/ (Accessed on February 1, 2023)

2.2.2. Unsupervised Learning

Unsupervised machine learning analyzes and groups unlabeled datasets using machine learning algorithms. ML algorithms identify hidden patterns or data clusters without the involvement of a human. The machine searches through a vast quantity of data to uncover insightful information. Additionally, dimensionality reduction is used to lower the number of features in a model. Algorithms for unsupervised machine learning don't need labels on the input data. There is no predefined outcome in unsupervised learning. The objective is to reorganize the incoming data into fresh features or a collection of objects with related patterns. Example algorithms include *the Apriori algorithm* and *K-Means*.

- **K-means**

  By reducing the variance within each cluster, the method repeatedly separates the data points into K clusters.
  Each data point is first allocated at random to one of the K clusters. Then, each data point is redistributed to the cluster with the closest centroid after computing each cluster's centroid (which is essentially its center). This procedure is repeated until there is no longer any change in the cluster assignments for any given data point.[10]

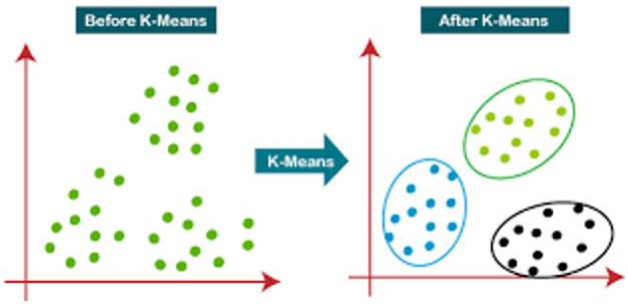


Figure 8. Before and after K-means implementation.
https://www.javatpoint.com/k-means-clustering-algorithm-in-machine-learning

- **KNN algorithm (K-nearest neighbours)**

  It is predicated on the notion that the observations most "similar" to a particular data point are those closest to it in the data set, allowing us to categorize unanticipated

[10] Python Machine Learning - K-means. (n.d.). Python Machine Learning - K-means. https://www.w3schools.com/python/python_ml_k-means.asp (Accessed on February 1, 2023)

points based on the values of the existing points that are closest to it in the data set, allowing us to categorize unanticipated points based on the values of the existing points that are closest to them. The user can choose K to specify how many neighbouring observations will be used in the algorithm.

The chosen number of closest neighbours is K. The class into which a new observation should be classified is decided by a majority vote. In comparison to very small values of K, larger values of K are frequently more resilient to outliers and provide more solid decision boundaries (K=3 would be preferable to K=1, which may yield unacceptable outcomes).[11]

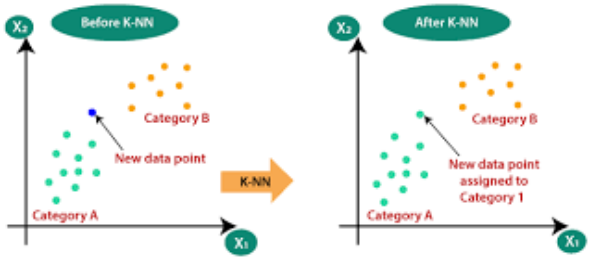


Figure 9. Before and after K-NN implementation. https://www.javatpoint.com/k-nearest-neighbor-algorithm-for-machine-learning

- **Neural Networks**

  An artificial intelligence technique called a neural network instructs computers to analyze data in a manner modeled after the human brain. Deep learning is a form of machine learning that employs linked neurons or nodes in a layered framework to mimic the human brain. It develops an adaptive system that computers utilize to continually learn from their errors and improve. Artificial neural networks so make an effort to accurately tackle complex issues, such as summarizing papers or identifying faces.[12]

[11] K-Nearest Neighbor(KNN) Algorithm for Machine Learning - Javatpoint. (n.d.). www.javatpoint.com. https://www.javatpoint.com/k-nearest-neighbor-algorithm-for-machine-learning (Accessed on February 1, 2023)
[12] What is a Neural Network? - Artificial Neural Network Explained - AWS. (n.d.). Amazon Web Services, Inc. https://aws.amazon.com/what-is/neural-network/ (Accessed on February 1, 2023)

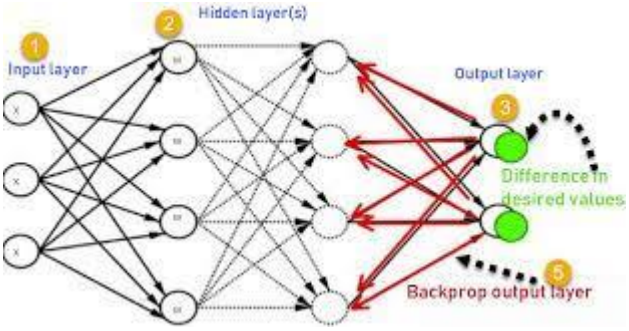


Figure 10. Neural Network implementation visual description.
https://www.guru99.com/backpropogation-neural-network.html

- **Hierarchal clustering**

  It is a machine learning algorithm that has the ability to make conclusions from a given dataset on its own, without the involvement of any human. A population or set of data points is divided into groups by a process called **clustering**, which makes the groups' members more similar to one another and distinct from those of other groups. In essence, it is a classification of things based on how similar and distinct they are.[13]

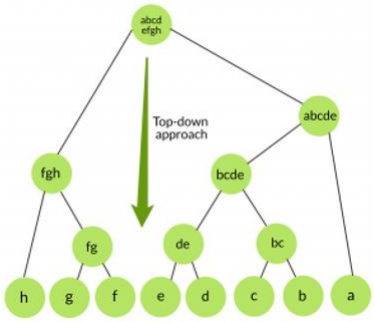


Figure 11. Hierarchal clustering implementation visual description.
https://www.geeksforgeeks.org/ml-hierarchical-clustering-agglomerative-and-divisive-clustering/

- **Anomaly detection**

  It is the method for detecting rare conditions or interpretations which can cause suspicions by being statistically different from the rest of the opinions. Such

[13] Mishra, U., & A. (2021, June 15). What is Hierarchical Clustering in Machine Learning? | Analytics Steps. What Is Hierarchical Clustering in Machine Learning? | Analytics Steps. https://www.analyticssteps.com/blogs/what-hierarchical-clustering-machine-learning (Accessed on February 1, 2023)

"anomalous" activity is often indicative of some sort of issue, such as credit card theft, a malfunctioning server component, a cyberattack, etc.[14]



Figure 12. Anomaly detection visual description.
https://www.geeksforgeeks.org/ml-hierarchical-clustering-agglomerative-and-divisive-clustering/

2.2.3. Semi-supervised Learning

Data scientists input a limited quantity of labeled training data to an algorithm to perform *semi-supervised learning*. The algorithm gains knowledge of the data set's dimensions from this and applies this knowledge to fresh, unlabeled data. The lack of sufficient labeled data for a supervised learning system can be resolved via semi-supervised learning. It also helps if labeling enough data is too expensive. When algorithms train on labeled data sets, their performance often becomes better. However, classifying data may be costly and time-consuming. The performance of supervised learning and the effectiveness of unsupervised learning are both met by semi-supervised learning. Example algorithms are extensions to other flexible methods that make assumptions about how to model the unlabeled data.

2.2.4. Reinforcement Learning

A learning agent in a *reinforcement learning* system receives a reward for each correct action and receives a penalty for each incorrect activity. With the help of this input, the agent automatically learns and becomes more effective. The agent explores and engages with the environment during reinforcement learning. An agent performs better since its objective is to increase the greatest reward points. Data scientists also train the algorithm to seek out rewards when it makes decisions that are beneficial toward the end objective and to avoid penalties when it takes actions that push it even further away from that end goal.

[14] Machine Learning for Anomaly Detection - GeeksforGeeks. (2019, May 27). GeeksforGeeks. https://www.geeksforgeeks.org/machine-learning-for-anomaly-detection/ (Accessed on February 1, 2023)

- **Q-Learning**

  The value of a given action in a particular scenario is determined using a model-free reinforcement learning method called Q-learning. Model-free refers to the fact that it can handle the stochastic transition and reward problems without requiring adjustments and that it is not reliant on a model of the environment.[15]

  Q-learning identifies an optimal strategy for any finite Markov decision process (FMDP) by optimizing the predicted value of the total reward across all subsequent steps, beginning from the present state. The most effective action-selection approach for any FMDP may be discovered using Q-learning with unlimited exploration time and a fairly random policy. The predicted rewards for a certain activity in a particular state make up the function "Q" that the approach computes.

$$Q^{new}(s_t, a_t) \leftarrow \underbrace{Q(s_t, a_t)}_{\text{current value}} + \underbrace{\alpha}_{\text{learning rate}} \cdot \overbrace{\Big(\underbrace{r_t}_{\text{reward}} + \underbrace{\gamma}_{\text{discount factor}} \cdot \underbrace{\max_a Q(s_{t+1}, a)}_{\text{estimate of optimal future value}} - \underbrace{Q(s_t, a_t)}_{\text{current value}} \Big)}^{\text{temporal difference}}$$
$$\underbrace{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad}_{\text{new value (temporal difference target)}}$$

Figure 13. Q-learning algorithm. https://en.wikipedia.org/wiki/Q-learning

- **Temporal Difference (TD)**

  Essentially, it is a technique for learning to predict a quantity that is dependent on the possible values of a certain signal. Using this method, the long-term utility of a pattern of behaviour is calculated using a series of intermediate incentives.
  In its most basic form, Temporal Difference Learning (TD Learning) focuses on predicting a variable's value in the future given a set of states. Thanks to temporal difference learning, the problem of reward prediction was eventually overcome. It may be argued that it employs a mathematical approach to replace sophisticated thought with a simple learning process that yields the same results.[16]

$$V(S_t) \leftarrow V(S_t) + \alpha \Big[R_{t+1} + \gamma V(S_{t+1}) - V(S_t) \Big]$$

Figure 14. Temporal Difference. https://towardsdatascience.com/introduction-to-reinforcement-learning-rl-part-6-temporal-difference-td-learning-2a12f0aba9f9

[15] Q-learning - Wikipedia. (2018, April 6). Q-learning - Wikipedia. https://en.wikipedia.org/wiki/Q-learning (Accessed on February 2, 2023)
[16] Temporal difference learning (TD Learning) | Engati. (n.d.). Engati. https://www.engati.com/glossary/temporal-difference-learning (Accessed on February 2, 2023)

2.3. The implementation of Machine Learning

The practical application of machine learning refers to the use of statistical models and algorithms to allow a machine or system to automatically improve its performance on a given job without being explicitly programmed. Making a machine learning algorithm or model workable so that it can be applied to addressing issues in the real world is what this approach entails. In this scenario, a computer is trained on a sizable dataset to discover correlations and patterns within the data, after which it applies what it has learned to forecast or decide on fresh, unforeseen data. Choosing an acceptable algorithm or model, gathering and preparing the data, training the machine, and assessing its performance are all phases of the implementation process. Building systems that can continuously and autonomously improve and enhance their performance without the assistance of a person is the ultimate objective of machine learning applications. When the machine learning model is put into practice, it may then be used to predict the future or carry out automated activities including finding patterns in data, making quick decisions, and coming up with new ideas. In plain English, machine learning implementation is the process of putting a theoretical idea into practice and having it function well in a particular setting.[17]

Examples of implementation of Machine Learning in day-to-day life:

1. **Image Recognition** - used in systems like facial recognition, self-driving cars, and security systems to recognize and classify objects in photos and videos.
2. **Fraud detection** – used to spot probable fraud instances immediately and detect unusual trends in financial activities. To identify fraudulent transactions, banks, and other financial institutions employ machine learning algorithms. The algorithms look for trends in the data to discover transactions that are out of the ordinary.
3. **Healthcare** - to forecast patient outcomes, identify illnesses, and suggest therapies, machine learning algorithms are employed in medical imaging, drug development, and personalized medicine. They can be used to evaluate medical pictures like X-rays or MRI scans to detect illnesses or ailments.

[17] Nordhausen, K. (2009, December). The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Second Edition by Trevor Hastie, Robert Tibshirani, Jerome Friedman. International Statistical Review, 77(3), 482–482. https://doi.org/10.1111/j.1751-5823.2009.00095_18.x (Accessed on February 1, 2023)

4. **Recommender Systems** - Online platforms like Amazon, Netflix, and Spotify use machine learning algorithms to recommend products, movies, and songs to their users based on their previous interactions and preferences.
5. **Predictive Maintenance** - Machine learning is used in predictive maintenance to evaluate data from industrial equipment and forecast when the repair is necessary, hence reducing downtime. By doing this, equipment dependability and safety are increased while downtime and maintenance costs are decreased.
6. **Natural Language Processing** - NLP applications including sentiment analysis, chatbots, language translation, and text categorization all involve machine learning techniques. These formulas enable computers to comprehend, translate, and produce human language.
7. **Speech Recognition** – ML algorithms are used in speech recognition systems, such as Siri, Alexa, and Google Assistant. These algorithms help computers transcribe and interpret spoken words accurately.
8. **Autonomous Driving** - Autonomous cars draw conclusions like steering, accelerating, and braking using ML algorithms. Real-time environmental forecasts and driving decisions are made after analysis of data from sensors, cameras, and other sources.

Chapter 3 - Cryptography and Cryptanalysis

3.1. What is Cryptography?

Cryptography is a field concerned with securing information and communication by transforming it into an encrypted text that only authorized parties can understand and access. The term is derived from "crypt" meaning hidden and "graphy" meaning writing. Cryptographic techniques use mathematical concepts and algorithms to protect information from unauthorized access. They are applied in various applications, such as key generation, digital signing, data privacy protection, secure web browsing, and protection of confidential transactions such as credit card and debit card transactions. The goal of cryptography is to establish secure communication in the presence of adversaries who aim to retrieve confidential information and compromise information security. In general, cryptography

focuses on developing and analyzing protocols to achieve confidentiality, integrity, authentication, and non-repudiation in communication.[18]

Examples of implementation of Cryptography in real-time:

- Data saved on a computer or delivered over the internet is encrypted to prevent unwanted access and eavesdropping by third parties.
- For the purpose of ensuring the legitimacy and integrity of documents, digital signatures are employed in electronic transactions and contracts.
- Online transactions and web browsing are protected by secure technologies like SSL and TLS.
- In a VPN, cryptographic key exchange protocols are utilized to securely exchange keys and create a secure connection.
- In order to check the accuracy of data, cryptographic hash functions are employed in software version control systems and digital forensics.
- Utilizing encrypted texting and secure phone communication in military and governmental activities.
- Mobile payments, internet banking, and other secure payment options that encrypt sensitive financial data including credit card transactions.

Below I will mention the features of Cryptography:

1. **Confidentiality**: to hide the content of the communication from unauthorized parties.
2. **Integrity**: to ensure that the information transmitted has not been altered during transmission.
3. **Authentication**: to verify the identity of the sender and receiver of the information.
4. **Non-repudiation**: to prevent the sender of the information from denying that they sent it.
5. **Key Management**: securely generate, store, and manage the cryptographic keys used for encryption and decryption.
6. **Scalability**: handle large amounts of data and provide security for various types of communication such as voice, text, and video.

[18] Cryptography Introduction - GeeksforGeeks. (2018, November 2). GeeksforGeeks. https://www.geeksforgeeks.org/cryptography-introduction/ (Acccessed on February 1, 2023)

7. **Flexibility**: adapt to changing security requirements and accommodate new technologies.
8. **Usability**: The ease with which cryptography can be used by end-users without requiring specialized knowledge or expertise.

3.1.1. Types of Cryptography[19]

- **Symmetric Key Cryptography**: It uses a single key for both encryption and decryption. This means that both the sender and the receiver of the message must have access to the same key in order to be able to encode and decode the information being sent. This type of encryption is considered to be faster and simpler than other forms of encryption, but there is a risk in that the key must be securely exchanged between the sender and receiver. Advanced Encryption Standard (AES), Blowfish, and Data Encryption Standard (DES) are examples of symmetric key cryptography.

- **Asymmetric Key Cryptography:** Also known as public key cryptography. It uses two keys, a public key for encryption and a private key for decryption. The public key is used to encrypt the data and can be freely distributed, while the private key is kept secret and is used to decrypt the data. This type of encryption is widely used in secure communication, such as online transactions, email, and file transfers.

- **Hash Function Cryptography:** Transforms an input (or "message") into a fixed-length output (or "hash") that acts as a digital fingerprint of the input using a mathematical operation called a "hash function." Numerous crucial characteristics of hash functions include the following:
    - Deterministic: Since the same input will always result in the same output, it is simple to check whether a message hasn't been changed.
    - Irreversible: The hash is perfect for use as a digital signature since it is computationally impossible to reconstruct the original message from the hash.
    - Unique: It is difficult for an attacker to generate a valid hash for a different message since even minor input changes result in totally distinct hashes.

[19] Cryptography and its Types - GeeksforGeeks. (2019, July 8). GeeksforGeeks. https://www.geeksforgeeks.org/cryptography-and-its-types/ (Accessed on February 1, 2023)

- **Block Cipher Cryptography**: It encrypts data one bit at a time and in fixed-size blocks as opposed to a stream cipher and is commonly used for data encryption. A symmetric key and algorithm are used by a block cipher to encrypt and decode a block of data. In order to enlarge the key space of the cipher and make it more challenging to crack the key via brute force, a block cipher needs an initialization vector (IV), which is appended to the input plaintext. The IV, which is obtained from a random number generator, is paired with the key, the first block's text, and the IV to make sure that the ciphertext produced by all subsequent blocks differs from the ciphertext produced by the first encryption block.[20]
- **Stream Cipher Cryptography**: A stream cipher is an encryption method that changes plain text into unintelligible code bytes by bytes, rendering it incomprehensible to nobody who has the right key. It encrypts data one bit or byte at a time and is commonly used for real-time encryption. Due to the linear nature of stream ciphers, the same key may be used to both encrypt and decode messages. Hackers have succeeded in breaking them even though it might be challenging. Because of this, experts believe stream ciphers should not be used widely. Nevertheless, a large number of individuals still rely on technology to transmit information through the internet. [21]

- **Public Key Infrastructure (PKI) Cryptography**: It uses asymmetric key cryptography to securely exchange keys and digital certificates for authentication and data encryption.

- **Elliptic Curve Cryptography (ECC):** It is a newer form of asymmetric key cryptography that uses mathematical equations based on elliptic curves to generate key pairs.

[20] What is a block cipher? (2021, May 1). Security. https://www.techtarget.com/searchsecurity/definition/block-cipher (Accessed on February 13, 2023)
[21] Stream Cipher 101: Definition, Usage & Comparisons | Okta. (n.d.). Stream Cipher 101: Definition, Usage & Comparisons | Okta. https://www.okta.com/identity-101/stream-cipher/#:~:text=A%20stream%20cipher%20is%20an,both%20encrypts%20and%20decrypts%20messages. (Accessed on February 13, 2023)

3.2. What is Cryptanalysis?

Cryptanalysis is the study of techniques for decrypting data without having access to the secret key that was used to encrypt it. To uncover flaws and vulnerabilities that may be exploited to disclose the original message or the encryption key entails examining cryptographic systems, ciphers, and algorithms. Cryptanalysis seeks to comprehend how encrypted data may be decoded without knowing the secret key and to spot any flaws in cryptographic systems that can make an attack effective.

Classical and contemporary cryptanalysis are the two main subcategories of cryptanalysis. The study of straightforward substitution ciphers and transposition ciphers is the emphasis of classical cryptanalysis, but current cryptanalysis is primarily concerned with complicated systems, such as symmetric and asymmetric encryption techniques.

Mathematical analysis, statistical analysis, and brute force attacks are a few of the techniques utilized in cryptanalysis. By examining patterns in the encrypted data, cryptanalysts employ mathematical methods like linear and differential cryptanalysis to attempt to ascertain the encryption key. Additionally, patterns in the encrypted data may be found and the encryption key can be discovered using statistical techniques like frequency analysis. When previous approaches have failed, brute force attacks—which include trying every key until the right one is found—are frequently employed as a final option.

Since the beginning of time, ciphers and codes have been employed to safeguard information and maintain confidentiality, giving rise to a long and rich history of cryptanalysis. Cryptanalysis has been employed historically for a wide range of tasks, including military intelligence, diplomatic correspondence, and criminal investigations.

Here are a few significant historical occurrences and milestones in the discipline of cryptanalysis:[22]

1. Simple substitution ciphers were employed in **ancient Greece and Rome** to encrypt messages and safeguard data. Additionally, during this time, cryptanalysis was utilized to understand enemy communications and acquire tactical benefits.

[22] The Code Book: The Secrets Behind Codebreaking. (n.d.). Google Books. https://books.google.com/books/about/The_Code_Book_The_Secrets_Behind_Codebre.html?hl=pl&id=8CHcJQzDPCkC (Accessed on February 13, 2023)

2. **World War I**: The employment of cryptanalysis by both the Allies and the Central Powers to intercept and decode enemy communications had a significant impact on how the war turned out.
3. The deciphering of the German Enigma code at Bletchley Park by British cryptanalysts, notably Alan Turing, during **World War II** is the most well-known instance of cryptanalysis in modern history. It is widely recognized as one of the most important accomplishments in the world of cryptography and played a crucial role in Allied triumphs.
4. Cryptanalysis continued to be important throughout the **Cold War** when the US and the Soviet Union fought a never-ending war to intercept and decode one other's communications.
5. **Modern era**: As encryption technologies have developed and advanced, cryptanalysis has recently grown in significance in the realms of cybersecurity and national security.

The complexity and sophistication of encryption methods mean that cryptanalysis is still a vital area in the twenty-first century. Cybersecurity, national security, and digital forensics are just a few of the modern-day uses for cryptanalysis.

The following are some instances of cryptanalysis being used in the twenty-first century:[23]

- **Cybersecurity**: To assess the robustness of encryption methods and identify flaws that attackers may exploit, cryptanalysis is employed. The security of systems and networks is enhanced by the development of better encryption techniques using this information.
- **National security**: Intelligence services and military groups utilize cryptanalysis to collect and decipher encrypted communications from potential enemies abroad.
- **Digital forensics**: During criminal investigations, encrypted data from confiscated devices like computers and smartphones is recovered via cryptanalysis.
- **Cryptocurrency**: Cryptanalysis is a technique for analyzing the security of cryptographic systems, such as blockchain technology and the encryption algorithms that protect financial transactions.

[23] Cryptography Engineering. (n.d.). Google Books. https://books.google.com/books/about/Cryptography_Engineering.html?hl=pl&id=nnqdQsrZnJgC (Accessed on February 13, 2023)

3.3. Ciphers

A cipher is a technique for encrypting a communication to protect its contents. The original communication is converted into an encoded message, or ciphertext, via ciphers using mathematical procedures. A secret key and knowledge of the encryption technique are needed to decrypt the ciphertext. To undo the encryption and get the original message, use the secret key.

Symmetric and asymmetric ciphers are the two primary categories into which ciphers may be split. The same key is used for both encryption and decryption in symmetric ciphers. Asymmetric ciphers often called public-key ciphers, employ distinct keys for encryption and decryption.

The oldest recorded uses of ciphers may be found in prehistoric societies, and the technology has a long history. The requirement for secure communication in a variety of circumstances, including military and diplomatic communications, has fueled the development of cryptography and ciphers in particular.

Modern cryptographic systems including secure communication protocols, digital signatures, and secure storage systems all require ciphers. They are essential in preventing unwanted access to and tampering with sensitive data, including financial transactions and personal data.

3.3.1. Cryptographic Machines [24]

Throughout history, communications have been encrypted and decrypted using cryptographic devices. Here are a few significant historical instances of cryptographic machines:

The Enigma Machine: During World War II, the German military used the Enigma machine, a mechanical encryption system, for secure communication. It was created to encrypt communications using an intricate system of rotors and wire, making it challenging to crack. A keyboard for entering plaintext, a display for presenting ciphertext, and a system of rotors and wire for converting the plaintext into ciphertext made up the device. The

[24] The Secret War. (n.d.). Google Books. https://books.google.com/books/about/The_Secret_War.html?hl=pl&id=VxRhCgAAQBAJ (Accessed on February 14, 2023)

Enigma machine employed a complicated system of rotors to modify how the alphabet's letters were mapped throughout the encryption process. Each message's encryption key was generated by a distinct configuration of the rotors, which could be adjusted to different beginning locations. The German military made extensive use of the device, which was regarded as being quite secure.

The efforts of Polish cryptanalysts before the war and British cryptanalysts during the war, including Alan Turing and his team at Bletchley Park, who worked tirelessly to understand the Enigma machine and develop methods for cracking the code, were ultimately successful and helped turn the tide of the war.

The SIGABA: During World War II, the US military employed the SIGABA, an electromechanical encryption machine. It was widely utilized by the US military and its allies for secret communication since it was thought to be one of the most secure cipher machines of its day.

To encrypt communications, the SIGABA utilized both electrical and mechanical rotors. The SIGABA cipher machine, unlike other cipher machines of the era, had a system of non-repeating rotor stepping and a sophisticated wiring scheme that made it exceptionally challenging to crack.

During the war, the SIGABA protected delicate military communications extremely well and was never successfully breached by the enemy. Its mechanical intricacy made it challenging to reverse engineer, and its employment of sophisticated cryptographic algorithms were major contributors to its security.

The Typex: During World War II, the Typex was a British encryption device. For secure communication between the British military and its allies, a modified version of the German Enigma machine was employed.

Since the Typex used many rotors to encrypt communications, it was considered to be exceptionally secure. The British military and its allies made extensive use of it during the war, and the enemy was never able to successfully topple it.

The Typex was employed to encrypt and decode delicate information pertaining to military operations, intelligence collection, and diplomatic communications. It was a crucial component of the British effort to win the war through intelligence gathering.[25]

The Lorenz SZ40/42: During World War II, the German military employed the Lorenz SZ40/42 teleprinter encryption equipment. High-level communication was conducted through it, and it was regarded as being quite secure. The device was challenging to crack because it employed a system of rotors and wires to encrypt information.

The Allies finally succeeded in breaking the code, which improved the war's outcome. The success of the Allies' intelligence-collecting activities throughout the war was greatly influenced by the breakthrough in cracking the Lorenz code, which was a huge achievement for them.

3.4. Decryption techniques[26]

Decryption is the process of converting encrypted or ciphertext back into its original form, called plaintext. There are several techniques used for decryption, including:

Brute force attack: This method involves trying every possible key until the correct one is found. This technique is often used when the encryption algorithm is known, but the key is unknown.

Known plaintext attack: This technique involves using knowledge of some of the plaintext and the corresponding ciphertext to determine the key.

Ciphertext-only attack: This technique involves using only the ciphertext to try and determine the key and the plaintext. This technique is used when there is no knowledge of the plaintext or the encryption algorithm.

[25] Cryptography: A Very Short Introduction. (n.d.). Google Books. https://books.google.com/books/about/Cryptography_A_Very_Short_Introduction.html?hl=pl&id=HvR0DgAAQBAJ (Accessed on February 14, 2023)
[26] Handbook of Applied Cryptography. (n.d.). Google Books. https://books.google.com/books/about/Handbook_of_Applied_Cryptography.html?hl=pl&id=YyCyDwAAQBAJ (Accessed on February 14, 2023)

Chosen plaintext attack: This technique involves choosing specific plaintext and observing the resulting ciphertext in order to determine the key.

Differential cryptanalysis: This technique involves analyzing the differences between ciphertexts to determine the key.

Linear cryptanalysis: This technique involves analyzing the linear relationships between ciphertexts and plaintexts to determine the key.

Side-channel attack: This technique involves using information from the encryption process, such as power consumption or electromagnetic emissions, to determine the key.

Chapter 4 – Integration of Machine Learning Algorithms and Cryptanalysis

4.1. Which Machine Learning Algorithms are used in Cryptanalysis?[27]

By examining their flaws and vulnerabilities, cryptanalysis is the process of dismantling cryptographic systems. Machine learning algorithms may be applied to cryptanalysis to find trends and learn information that can help crack these systems. Examples of machine learning algorithms used in cryptanalysis include the following:[28]

- **Decision trees:** Message encryption keys are predicted using decision trees, which examine ciphertexts. By examining well-known ciphertexts and the plaintexts that go with them, the tree is constructed. The key used to encrypt a new ciphertext is then predicted using this data.
- **Neural networks:** Neural networks can be used to break various cryptographic systems, including substitution ciphers and transposition ciphers. They work by analyzing a large number of ciphertexts and their corresponding plaintexts and then using this information to predict the plaintext corresponding to a new ciphertext.

[27] Kocarev, L., & Zukerman, S. L. (2012). Machine learning in cryptography. IEEE Transactions on Neural Networks and Learning Systems, 23(1), 3-13.
[28] Fouché, P., Tomaselli, M. V., & Neto, R. F. (2013). Cryptanalysis of symmetric-key cryptographic systems using machine learning techniques. Journal of Cryptographic Engineering, 3(1), 29-45.

- **Genetic algorithms:** To get the best key for a specific encryption technique, genetic algorithms are utilized. They function by creating a population of keys, which are then evaluated for quality using a fitness function. The procedure is repeated until an appropriate key is discovered, selecting the best keys for the subsequent generation.
- **Support vector machines (SVMs):** Anomalies in cryptographic systems can be found using SVMs. They operate by examining the patterns in known ciphertexts and utilizing this knowledge to spot unusual ciphertexts that could point to a flaw in the encryption technique.

4.2. The implementation techniques and use cases[29]

Key recovery, ciphertext-only attacks, and side-channel attacks are only a few of the many uses for machine learning techniques in cryptanalysis. Here are some instances of machine learning algorithms' application strategies and use cases in cryptanalysis:

- **Decision trees for key recovery:** The Advanced Encryption Standard (AES) and the Data Encryption Standard (DES) are two cryptographic methods that have utilized decision trees for key recovery. AES keys were successfully recovered from ciphertexts using decision trees in research by Zhang et al. (2018), with a success rate of over 98%.[30]
- **Neural networks for ciphertext-only attacks**: Numerous classical ciphers, including the Caesar, Vigenère, and Playfair ciphers, have been cracked using neural networks. The Vigenère encryption was cracked using a neural network in a research by Baccouche et al. (2012), with a success rate of over 80% from ciphertext-only assaults.[31]
- **Support vector machines for side-channel attacks:** In order to crack cryptographic systems, SVMs have been employed to assess side-channel data such as power use

[29] Farhat, A., Allam, L., & El Ghazi, H. (2015, July). Cryptanalysis of classical ciphers based on genetic algorithms. In Proceedings of the International Conference on Security and Cryptography (pp. 1-6).
[30] Zhang, S., Li, X., Li, T., Zhou, Y., & Li, Q. (2018). A decision tree based attack on AES. Journal of Computer and System Sciences, 88, 209-220.
[31] Baccouche, M., Mamalet, F., Wolf, C., Garcia, C., & Brouard, T. (2012). Sequential deep learning for human action recognition. In Proceedings of the 2012 International Conference on Human-Computer Interaction (pp. 29-39).

and electromagnetic radiation. SVMs were employed in a study by Messerges et al. (1999) to evaluate power usage metrics and crack DES keys.[32]

- **Deep learning for automated cryptanalysis**: Convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are examples of deep learning algorithms that have been applied to automated cryptanalysis, which aims to crack cryptographic systems automatically and without the assistance of a person. CNNs were used to decipher monoalphabetic substitution ciphers in research by Abadi and Zimmermann (2014).[33]

Chapter 5 – The "Decrypted" project

5.1. General idea and approach to the topic[34] [35] [36]

In this chapter I would like to illustrate the implementation of machine learning algorithms in cryptanalysis from my perspective by a use case. My project is inspired by a research paper called "Sequential Deep Learning for Human Motion Recognition" presented by Baccouche et al. (2012) which discusses the use of deep learning algorithms for human action recognition in video sequences. The authors propose a method that uses deep neural networks (DNNs) to extract features from video frames, and a hidden Markov model (HMM) to model the temporal dynamics of the action sequence.

The authors begin by discussing the limitations of existing methods for human action recognition, which typically use hand-crafted features and suffer from limited discriminative power and sensitivity to variations in the input data. They then describe their proposed

[32] Messerges, T. S., Dabbish, E. A., & Sloan, R. H. (1999). Examining smart-card security under the threat of power analysis attacks. IEEE Transactions on Computers, 48(9), 1010-1018.
[33] Abadi, M., & Zimmermann, R. (2014). Machine learning for cryptanalysis. In Proceedings of the 5th Conference on Security and Cryptography for Networks (pp. 1-8).
[34] Baccouche, M., Mamalet, F., Wolf, C., Garcia, C., & Baskurt, A. (2012). Sequential deep learning for human action recognition. In Proceedings of the International Workshop on Human Behavior Understanding (pp. 29-39). Springer.
[35] Baccouche, M., Mamalet, F., Wolf, C., Garcia, C., & Baskurt, A. (2012). Sequential deep learning for cryptographic ciphertext classification. International Conference on Structural, Syntactic, and Statistical Pattern Recognition, 18-29.
[36] Baccouche, M., Mamalet, F., Wolf, C., Garcia, C., & Baskurt, A. (2012). Sequential deep learning for Vigenère cipher breaking. In Proceedings of the 21st International Conference on Pattern Recognition (ICPR2012) (pp. 1617-1620). IEEE.

method, which uses a DNN to extract hierarchical features from the video frames, and an HMM to model the temporal dynamics of the action sequence. The authors show that their method outperforms state-of-the-art methods on several benchmark datasets, demonstrating the effectiveness of deep learning algorithms for action recognition.

The paper provides a detailed description of the proposed method, including the architecture of the DNN and the HMM, and the training and testing procedures. The authors also provide a comprehensive evaluation of their method, comparing it to several state-of-the-art methods on several benchmark datasets, and analyzing the impact of various factors on its performance.

Overall, Baccouche et al. (2012) provides valuable insights into the use of deep learning algorithms for human action recognition, demonstrating the effectiveness of these algorithms for complex and challenging tasks in computer vision.

In my version of project, the code is an implementation of the Vigenère cipher. The program generates a random plaintext and key, encrypts the plaintext using the Vigenère cipher, and then uses a neural network to perform a frequency analysis attack on the ciphertext and recover the original plaintext. The detailed description and documentation are mentioned in chapter 5.2.

The Vigenère cipher is a classical encryption method that uses a keyword to encrypt a message. The keyword is repeated over the plaintext, and each letter of the plaintext is shifted according to the corresponding letter of the keyword. The result is the ciphertext, which can be decrypted using the same keyword. However, it should be noted that the Vigenère cipher is a relatively weak cipher that can be easily broken using frequency analysis, and should not be used for secure communication. This code is for educational purposes only and is not intended to be used for actual encryption or decryption. The purpose is to demonstrate the principles of using machine learning techniques to attack cryptographic ciphers. In order to achieve my goal I used Python libraries and Neural Networks training model.

5.2. How to use it?

In this section, I would like to explain how the project works.

1. As a first step I imported the required library in order to work with my project which is one of the Python libraries called “NumPy”.

```
import numpy as np
```

Figure 15. A section from the project that is created by the author of this thesis.

2. Then followed with a section that generates a random plaintext and key (which have mentioned examples that can be used for testing purposes are commented nearby) and uses the key to encrypt the plaintext using the Vigenère cipher. In order to achieve this, I defined a function **vigenere_encipher** to encrypt the plaintext using the Vigenère cipher.

```
# generate some random plaintext and key
plaintext = input("Enter a plaintext: ") #'I have it all done'
key = input("Enter a key: ")#'secret'

# encrypt the plaintext using the Vigenère cipher
def vigenere_encipher(plaintext, key):
    ciphertext = ''
    for i in range(len(plaintext)):
        pi = ord(plaintext[i]) - 97
        ki = ord(key[i % len(key)]) - 97
        ci = (pi + ki) % 26
        ciphertext += chr(ci + 97)
    return ciphertext

ciphertext = vigenere_encipher(plaintext, key)
```

Figure 16. A section from the project that is created by the author of this thesis.

3. Next, I illustrated two functions **gen_feature_vectors** and **gen_labels** to generate the feature vectors and labels needed for training the neural network. The feature vectors are generated by selecting substrings of length **m** from the ciphertext and counting the frequency of each letter in the substring. **gen_feature_vectors** takes the ciphertext and a parameter **m** as input, and returns a list of feature vectors of length **m** for each possible substring of length **m** in the ciphertext. Each feature vector is a one-hot encoding of the frequencies of the letters in the substring. **gen_labels** takes the ciphertext and **m** as input, and returns a list of labels, which are the indices of the letters following each substring in the ciphertext. The training data is split into 80% training and 20% testing data.

```python
# generate the feature vectors
def gen_feature_vectors(ciphertext, m):
    feature_vectors = []
    for i in range(len(ciphertext) - m):
        feature_vector = [0] * 26
        for j in range(m):
            index = ord(ciphertext[i + j]) - 97
            feature_vector[index] += 1
        feature_vectors.append(feature_vector)
    return feature_vectors

# generate the labels
def gen_labels(ciphertext, m):
    labels = []
    for i in range(len(ciphertext) - m):
        labels.append(ord(ciphertext[i + m]) - 97)
    return labels
```

Figure 17. A section from the project that is created by the author of this thesis.

4. The below mention code trains a multi-layer perceptron (MLP) neural network on the training set using **scikit-learn**'s **MLPClassifier** class. The neural network has two hidden layers, each with 100 neurons. I aim to describe this piece of code by generating a feature vector from the last **m** characters in the ciphertext, predicting the label using the MLP classifier, and using the predicted label to determine the next character in the plaintext. This process is repeated until the entire plaintext is recovered.

   Then I generated the training and testing da by calling **gen_feature_vectors** and **gen_labels** on the ciphertext and using the first 80% of the resulting feature vectors and labels as the training set, and the remaining 20% as the testing set.

```
# generate the labels
def gen_labels(ciphertext, m):
    labels = []
    for i in range(len(ciphertext) - m):
        labels.append(ord(ciphertext[i + m]) - 97)
    return labels

# generate the training and testing data
m = 6
feature_vectors = np.array(gen_feature_vectors(ciphertext, m))
labels = np.array(gen_labels(ciphertext, m))

split = int(0.8 * len(feature_vectors))
train_X = feature_vectors[:split]
train_y = labels[:split]
test_X = feature_vectors[split:]
test_y = labels[split:]

# train the model
from sklearn.neural_network import MLPClassifier

model = MLPClassifier(hidden_layer_sizes=(100, 100))
model.fit(train_X, train_y)
```

Figure 18. A section from the project that is created by the author of this thesis.

5. To predict the next character in the plaintext given the current ciphertext, key, and neural network I defined a function **predict_next_character**. The function takes the ciphertext, key, **m**, and the neural network as input, and returns the updated ciphertext and key. Then it uses the **predict_next_character** function to iteratively predict the next character in the plaintext until the entire plaintext has been recovered.

```
# predict the next character
def predict_next_character(ciphertext, key, m, model):
    feature_vector = gen_feature_vectors(ciphertext, m)[-1]
    label = model.predict([feature_vector])[0]
    next_char = chr((label + 97) % 26 + 97)
    key += next_char
    ciphertext += next_char
    return ciphertext, key

# decrypt the ciphertext
key = ''
while len(key) < len(plaintext):
    ciphertext, key = predict_next_character(ciphertext, key, m, model)

print(f'ciphertext: {ciphertext}')
print(f'key: {key}')
```

Figure 19. A section from the project that is created by the author of this thesis.

6. The output includes the decrypted plaintext and the key used for encryption.

Chapter 6 – Conclusion

6.1. Future implementations

In recent years, cryptanalysis using machine learning algorithms has produced encouraging results. Machine learning algorithms will probably become more and more crucial as the subject of cryptography develops, making it easier to crack codes and create new encryption techniques. The application of machine learning in cryptanalysis, however, may face difficulties in the future due to the development of increasingly complex encryption schemes and the growing accessibility of quantum computing. Additionally, ethical issues including privacy worries and possible exploitation of these algorithms must be addressed. Overall, the future of using machine learning algorithms to cryptanalysis will depend on ongoing research and development in both cryptography and machine learning.

6.2. Summary

In conclusion, machine learning algorithms have shown a lot of promise in the domain of cryptanalysis, especially in cracking ciphers. Prior to the development of machine learning techniques, the Vigenère encryption was thought to be impenetrable. Because they can spot trends in encrypted data, evaluate those patterns, and then utilize them to assist crack ciphers, machine learning algorithms are extremely helpful in cryptanalysis. Being able to discern intricate patterns in huge datasets makes neural networks particularly useful in this area.

Some of the early machine learning approaches were employed to decipher codes during World War II, demonstrating the long history of machine learning algorithms in cryptanalysis. Machine learning algorithms are being employed in fields including financial fraud detection, speech recognition, and picture identification in order to assist safeguard computer systems from cyberattacks.

Overall, the use of machine learning algorithms for cryptanalysis is a fast developing subject where new discoveries are being produced every day. We can anticipate more advanced machine learning algorithms that can decipher even the most difficult ciphers being created as technology progresses.

Images

Figure 1. Linear regression visual description. https://www.javatpoint.com/linear-regression-vs-logistic-regression-in-machine-learning

Figure 2. Logistic regression visual description. https://www.javatpoint.com/linear-regression-vs-logistic-regression-in-machine-learning

Figure 3. Decision tree visual description. https://www.javatpoint.com/machine-learning-decision-tree-classification-algorithm

Figure 4. SVM algorithm visual representation. https://www.javatpoint.com/machine-learning-support-vector-machine-algorithm

Figure 5. Naïve Bayes theorem. https://www.naukri.com/learning/articles/naive-bayes-algorithm-in-machine-learning/#nba

Figure 6. Naïve Bayes theorem and algorithm visual representation. https://www.naukri.com/learning/articles/naive-bayes-algorithm-in-machine-learning/#nba

Figure 7. Random forest algorithm visual representation. https://www.tibco.com/reference-center/what-is-a-random-forest

Figure 8. Before and after K-means implementation. https://www.javatpoint.com/k-means-clustering-algorithm-in-machine-learning

Figure 9. Before and after K-NN implementation. https://www.javatpoint.com/k-nearest-neighbor-algorithm-for-machine-learning

Figure 10. Neural Network implementation visual description. https://www.guru99.com/backpropogation-neural-network.html

Figure 11. Hierarchal clustering implementation visual description. https://www.geeksforgeeks.org/ml-hierarchical-clustering-agglomerative-and-divisive-clustering/

Figure 12. Anomaly detection visual description. https://www.geeksforgeeks.org/ml-hierarchical-clustering-agglomerative-and-divisive-clustering/

Figure 13. Q-learning algorithm. https://en.wikipedia.org/wiki/Q-learning

Figure 14. Temporal Difference. https://towardsdatascience.com/introduction-to-reinforcement-learning-rl-part-6-temporal-difference-td-learning-2a12f0aba9f9

Figure 15. A section from the project that is created by the author of this thesis.

Figure 16. A section from the project that is created by the author of this thesis.

Figure 17. A section from the project that is created by the author of this thesis.

Figure 18. A section from the project that is created by the author of this thesis.

Figure 19. A section from the project that is created by the author of this thesis.

List of Abbreviations

IT – Information Technology

ML – Machine Learning

BPA – Business Process Automation

NLP – Natural Language Processing

SVM – Support Vector Machine

SSL – Secure Socket Layer

TLS – Transport Layer Security

VPN – Virtual Private Network

PKI – Public Key Infrastructure

ECC – Elliptic Curve Cryptography

AES – Advanced Encryption Standard

DES – Data Encryption Standard

CNN – Convolutional neural network

RNN – Recurrent neural network

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